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| **Fiscal Year:** | FY 2019 | **Task Last Updated:** | FY 07/27/2018 |
| **PI Name:** | Plawsky, Joel Sc.D. | | |
| **Project Title:** | Constrained Vapor Bubble Flight Experiment II – Binary Mixtures | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Fluid physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Troy | **State:** | NY |
| **Zip Code:** | 12180-3522 | **Congressional District:** | 20 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | NOT AVAILABLE |
| **Start Date:** | 10/01/2013 | **End Date:** | 12/31/2019 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Chao, David | **Contact Phone:** | 216-433-8320 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 12/31/2019 per NSSC information (Ed., 5/13/19)<br>NOTE: End date changed to 3/31/2019 per NSSC information (Ed., 12/7/17) | | |
| **Key Personnel Changes/Previous PI:** | July 2017 report: Graduate student Akshay Kundan graduated; Graduate student Thao Nguyen graduated; Graduate student Jiaheng Yu joined the group. July 2018 report: Jiaheng Yu is working on the project and Sarah Dalakos is an undergrad working for the summer. | | |
| **COI Name (Institution):** | Wayner, Peter Ph.D. ( Rensselaer Polytechnic Institute ) | | |
| **Grant/Contract No.:** | NNX13AQ78G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The thermophysical principles underlying change-of-phase heat transfer systems controlled by interfacial phenomena under microgravity conditions are not well understood. As a result, related passive engineering systems like heat pipes have not been optimized and it is unclear whether recent advances in the design of these systems based on ground-based research using fluid mixtures can be translated to microgravity systems. Herein, we propose basic experimental and theoretical studies of the nonisothermal constrained vapor bubble (CVB-2) using a mixture of pentane and isohexane under microgravity conditions to help remedy this undesirable situation. This proposed study is multi-faceted: 1) it is a basic scientific study in interfacial phenomena, microgravity fluid physics, and thermodynamics; 2) it is a basic study in thermal and mass transport; and 3) it is a study of a passive heat exchanger. Although the basic scientific Facets (1) and (2) are emphasized, the research naturally leads to the development of a passive heat exchanger optimized for microgravity conditions.<br>The proposed optical study of a vapor bubble constrained in a transparent glass cell under nonisothermal conditions will increase the basic understanding of transport systems controlled by interfacial phenomena. The pressure gradient field will be obtained optically whereas the temperature field will be obtained using thermal sensors. Due to the sensitivity of systems of the proposed relatively large size to gravity, to small temperature and pressure gradients, and to changes in fluid composition, these transport systems need to be studied under microgravity conditions. An "axisymmetric" system with a small Bond number is required. Direct extensions to the microgravity environment first investigated in CVB-1 are proposed. The augmented Young-Laplace equation and the Kelvin-Clapeyron model will be used to help understand and quantify how the system operates.<br><br>The immediate macroscopic objectives of CVB-2 are to determine the stability, the fluid flow characteristics, the average heat transfer coefficient in the evaporator and condenser sections, and the heat conductance of the CVB as a function of the heat input using the new fluid mixture. The immediate microscopic objective is to determine the detailed characteristics of the transport processes in the curved liquid film. The local conditions under which cavitation and instability occur with the formation of a dry region will be determined as a function of heat flux, film thickness, and stress. High fluxes and relatively large systems with small capillary pressures will be emphasized. The data using the pentane/isohexane mixture will be compared with the pure pentane data obtained in CVB-1 and ground-based data on identical fluid mixtures. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This work addresses fundamental questions regarding phase-change heat and mass transfer operations that are becoming increasingly important in a wide variety of fields ranging from energy conversion, to microelectronics cooling, micro-electromechanical systems (MEMS) devices, self-assembly, and space flight. Of particular concern to NASA are microchannel and heat pipe systems used for thermal management and perhaps separation processes. Such systems are attractive for long-term space flights because they generally require no moving parts and are potentially maintenance free.<br>The ability to control transport processes in an extended meniscus will be of great utility both on Earth and in space, whether we are looking for enhanced heat transfer, coating, or self-assembly applications. Some of our preliminary ground-based experiments concerning the microscopic evaporating meniscus, performed as part of the existing CVB-1 flight experiment program, have indicated that using a fluid mixture, with the minority component being the less volatile, higher surface tension liquid, offers the opportunity to control fluid flow and heat transfer in an evaporating meniscus and thus enhance the performance of heat pipes. The surface concentration gradient will also enhance axial fluid flow in the corners. The extension of the CVB-1 experiment to mixtures allows us to isolate the Marangoni effects from the effects of gravity and begin to understand how to exploit Marangoni flows for improving the performance of heat pipes in space and on Earth. |
| **Task Progress:** | 1. We have made great progress in furthering our understanding of what drives the fluid motion and vapor-liquid distribution inside the CVB.<br>2. We have analyzed the effect of cooler temperature on the evaporator performance of CVB-1. Even though the cooler is located nearly 40 mm away from the heater, altering the cooler temperature had a great effect on the heat pipe's performance and the strength of Marangoni drive flows. At the moment it appears that the length of the Marangoni region and the location of our junction point between capillary drive and Marangoni driven flows all depend on the surface tension gradient between the heater wall and the point at which condensation first starts.<br><br>3. We are still finishing up a paper that explains what happens in CVB-1 once the Marangoni region stops increasing in length. At that point, we see a jet of fluid along the flat portion of the wall. Our analysis tells us that it is akin to a rip current driven by counterrotating vortices present in the central drop. These act to channel fluid between them and spit that fluid out toward the heater. The end result is that we can maintain the heat transfer and evaporative capacity of the system by forming new vapor-liquid interfaces in the central region of the system instead of the corners. We are still working on resolving that paper and want to submit it to PNAS (Proceedings of the National Academy of Sciences of the United States of America).<br><br>4. Our project with Southern Methodist University (SMU) to use the Physical Sciences Informatics database to continue analyzing and modeling results from CVB1 was funded and we are working on advanced models to explain the phenomena we have observed.<br><br>5. We have rebuilt our experimental, ground-based capability and are now beginning to look at mixtures. We want to see if we get nonlinearity in Hamaker constant as Bill Meyer saw in surface tension. Then we will run mixtures under heating.<br><br>6. We presented results at 3 meetings in 2017, one in summer of 2018, and are scheduled for 3 more in Fall/Winter 2018.<br><br>7. Thao Nguyen defended her thesis in the Spring of 2017 and graduated in Fall of 2017.<br><br>8. We published a paper explaining why menisci spontaneously oscillate and what the advantage of that oscillation is. The oscillation needs to be violent enough so that condensation occurs during a portion of the cycle. This decreases the disjoining pressure and allows for increased evaporation. |

| Bibliography Type: | Description: (Last Updated: 10/14/2025) |
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| Abstracts for Journals and Proceedings | Nguyen TT, Yu J, Wayner PC Jr, Plawsky JL. "Modeling the Effect of Oscillation in an Evaporating Meniscus on the Heat Transfer Process." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017.<br>Abstracts. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017 |
| Abstracts for Journals and Proceedings | Nguyen TT, Wayner PC Jr, Plawsky JL. "Condensation on highly superheated surfaces: unstable thin films in a wickless heat pipe." 2017 AIChE National Meeting, Minneapolis, MN, October 29 - November 3, 2017.<br>Abstracts. 2017 AIChE National Meeting, Minneapolis, MN, October 29 - November 3, 2017. , Nov-2017 |
| Articles in Peer-reviewed Journals | Plawsky JL, Nguyen T. "Wickless heat pipes in microgravity." Physics Today. 2017 Sep;70(9):82-3. https://doi.org/10.1063/PT.3.3704 , Sep-2017 |
| Articles in Peer-reviewed Journals | Nguyen TT, Yu J, Plawsky JL, Wayner PC, Chao DF, Sicker RJ. "Spontaneously oscillating menisci: Maximizing evaporative heat transfer by inducing condensation." International Journal of Thermal Sciences. 2018 Jun;128:137-48. https://doi.org/10.1016/j.ijthermalsci.2018.02.015 , Jun-2018 |
| Awards | Plawsky JL. (Joel L Plawsky) "2017 ASME ICNMM (American Society of Chemical Engineers International Conference on Nanochannels, Microchannels, and Minichannels) Prominent Researcher Award, August 2017." Aug-2017 |